Bintang Atmajaya

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PROFILE

Fresh graduate in Industrial Mechanical Engineering (Energy Conversion) from Institut Teknologi Sepuluh Nopember (ITS) with a GPA of 3.24/4.00, specializing in industrial energy systems, fluid power, and mechanical asset maintenance. Possesses hands-on experience in power plant equipment maintenance through an overhaul internship at PLN Nusantara Power UMRO, focusing on gas turbine Combustion Inspection (CI) and component reliability. Holds BNSP certification in Pneumatic & Hydraulic Systems Maintenance, complemented by strong engineering design and simulation capabilities (Autodesk Inventor, AutoCAD, ANSYS FEA) applied to piping systems, centrifugal pumps, and structural optimization. Combines technical engineering analytical skills with project management and organizational leadership (PMO). Seeking entry-level opportunities as a Project Engineer, Maintenance Engineer, or Production Engineer.

EDUCATION

Institut Teknologi Sepuluh Nopember Surabaya

Bachelor of Applied Science (D4) in Industrial Mechanical Engineering (Energy Conversion Focus)
August 2022 – July 2026 | Cumulative GPA: 3.24/4.0
Thesis: Design of a Leg System for a Wall-Climbing Robot Based on Negative Pressure
Designed the mechanical structure and timing-belt drive system and performed structural analysis

INTERNSHIP EXPERIENCE

PLN Nusantara Power UMRO — Gresik, Indonesia
Mechanical Maintenance Intern – Overhaul Division | Feb 2025 – Jun 2025

- Executed gas turbine Combustion Inspection (CI) overhaul activities, including disassembly, technical inspection, and reassembly of combustion systems.
- Inspected critical components such as combustion liners, transition pieces, and fuel nozzles for wear and structural integrity following strict K3/HSE standards.
- Documented maintenance inspection data to support equipment reliability and preventive maintenance decision-making.

Bayu Caraka ITS Robotics Team — Surabaya, Indonesia
Mechanical Staff Intern – VTOL & Fixed-Wing Divisions | Aug 2024 – Sep 2024

- Assisted in designing frame structures and integrating propulsion mechanisms for VTOL and Fixed-Wing UAVs.
- Fabricated lightweight structural components using 3D printing (PLA+) and composite materials to maintain optimal strength-to-weight ratios.
- Performed pre-flight mechanical inspections, assembly, and rapid repairs during field flight testings.

KEY ENGINEERING PROJECTS

- Wall-Climbing Robot Chassis (Thesis Project): Designed a tracked-wheel robot chassis using negative pressure adhesion. Conducted motor sizing, FEA stress analysis, and topology optimization on 3D-printed PLA+ materials using Autodesk Inventor and ANSYS. Authored a research paper formatted for the POMITS journal.
- Centrifugal Pump & Piping System Design: Drafted Process Flow Diagrams (PFD) and Piping & Instrumentation Diagrams (P&ID) for a centrifugal pump installation using AutoCAD, including Bill of Quantities (BOQ) and technical specification reports.
- Shell-and-Tube Heat Exchanger Design: Designed a shell-and-tube heat exchanger for thermal energy recovery, performing heat balance calculations, Logarithmic Mean Temperature Difference (LMTD) analysis, surface area sizing, and pressure drop evaluation.
- Off-Grid Solar Photovoltaic (PLTS) System Design: Designed an off-grid solar photovoltaic (PLTS) system for renewable power generation, conducting daily load profiling, PV array and battery bank sizing, inverter selection, and energy output estimations.
- Automated Pneumatic Assembly Line Conveyor: Designed an automated material handling conveyor system for assembly line operations using electro-pneumatic actuators, directional control valves, and sequential circuit logic.
- Power Plant Combustion Overhaul Study: Evaluated combustion inspection workflows and component reliability during power plant maintenance activities at PLN NP UMRO.
- Automated Mosquito Repellent Device (POMN 2024): Designed and developed an automated device for the "Program Ormawa Membangun Negeri" community project in Watu Agung, Gresik.

ORGANIZATIONAL EXPERIENCE

Badan Eksekutif Mahasiswa Fakultas Vokasi ITS (BEM FV ITS) — Surabaya, Indonesia
Staff of Strategic Studies Division (Kastrat) | Mar 2024 – Dec 2025

- Conducted research and policy analysis regarding technology, social issues, and student development.
- Formulated strategic recommendations to support organizational decision-making and policy advocacy.

UKM BRIDGE ITS — Surabaya, Indonesia
Vice Head of External Affairs Department | Mar 2023 – Dec 2025

- Managed external institutional partnerships and strategic outreach with partner universities and external organizations.
- Coordinated external communications and delegation logistics for national-level student engagements.

CERTIFICATIONS, TRAINING & COMPETENCIES

BNSP Certified Maintenance of Pneumatic Automation Devices & Hydraulic Systems
November 6th 2025

- Competent in reading, interpreting, and analyzing complex pneumatic, electro-pneumatic, and hydraulic circuit schematics.
- Skilled in preventive and corrective maintenance of hydraulic power packs, pumps, air compressors, and fluid power actuators.
- Experienced in testing, adjusting, and troubleshooting directional control valves, pressure valves, and sensor-integrated automated systems.
- Proficient in systematic fault diagnosis, component isolation, and safe replacement of worn system components according to industrial SOPs.
- Strict adherence to occupational health, safety, and environmental (K3/HSE) protocols during fluid power and automation maintenance.

PMO (Pelatihan Manajerial Organisasi) – ITS | Apr 2024

- Trained in top-tier student organization governance, strategic leadership, and executive-level organizational management.

PKTI (Pelatihan Karya Tulis Ilmiah) – ITS | Jun 2023

- Completed training on scientific writing methodologies, research structuring, and academic paper development.

UME (Upgrading Managerial Event) – UKM BRIDGE ITS | Apr 2023

- Enhanced managerial capabilities in executing structured organizational workflows, event governance, and team coordination.

LKMW (Latihan Keterampilan Manajemen Wirausaha) – ITS | Mar 2023

- Completed entrepreneurship and business management training focused on venture planning and execution.

LKMM Pra-TD (Latihan Keterampilan Manajemen Mahasiswa) – ITS | Dec 2022

- Developed foundational managerial competencies including strategic decision-making, active/passive communication, and people management.

LEADERSHIP & COMMITTEE EXPERIENCE

- BITE (BRIDGE Invite x UNAIR 2023) — Project Officer / Event Chairperson | Nov 2023 – Dec 2023

Directed a cross-university event between ITS and Universitas Airlangga, overseeing overall event strategy, operations, budget allocation, and risk mitigation

- PAMA Goes to Campus 2024 — Liaison Officer (LO) | May 2024

Served as the corporate point of contact for PT Pamapersada Nusantara representatives, coordinating executive roadshow schedules and hospitality.

- Bursa Karir ITS (Series 43, 44, 46) — Operations & Logistics Committee | Jan 2024 – Apr 2025

Coordinated administration, equipment setup, vendor distribution, and consumption logistics for ITS's largest career fair series.

- Temu Ortu MABA ITS x IKOMA 2026 — Liaison Officer (LO) | Sep 2026

Managed VIP guest reception, protocol requirements, and operational flow for thousands of incoming student guardians during the university orientation assembly.

SKILLS & ACHIEVEMENTS

- Technical Skills: Autodesk Inventor, AutoCAD, AutoCAD Plant 3D, ANSYS (FEA & Topology Optimization), Piping System Design (PFD & P&ID), Centrifugal Pump Systems, Gas Turbine Overhaul (Combustion Inspection), Liquid Penetrant Testing (NDT), Industrial Pneumatics & Hydraulics (BNSP Standard), ESP32/Arduino, 3D Printing (PLA+)..
- Soft Skills: Project Management, Maintenance Planning, Public Speaking, Negotiation, Analytical Thinking, Team Leadership.
- Achievements: 4th Standing Champion – Futurest National Debate Competition (2024)
- Selected Grantee – Program Ormawa Membangun Negeri (POMN) 2024.

LANGUAGES

- Indonesia (Native)
- English (B2 / Upper-Intermediate)
- Spanish (Elementary / Basic)